SLIIT UNI
THE KNOWLEDGE UNIVERSITY

IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

Project ID: **KND_CDAP_IT_05**

1. Topic (12 words max)

Multi-Factor Student Attrition Risk Prediction Using Academic, Engagement, Psychosocial, Environmental Factors

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Student attrition remains a significant challenge within Sri Lankan higher education institutions. When a student withdraws or is academically terminated, the consequences extend beyond the individual to affect their family and the institution as a whole. Most early-warning systems currently in use only identify at-risk students after a noticeable decline in academic performance has already occurred (for instance, a GPA falling below 2.0), by which point there may be limited opportunity for meaningful intervention.

This problem also affects students unequally: those from lower-income or rural backgrounds tend to be disproportionately impacted, as most existing systems rely solely on academic indicators and disregard financial or environmental stressors that may be contributing to their risk.

This project falls under the CoEAI research cluster, applying explainable AI techniques to the field of education. It aligns with SDG 4 (Quality Education) by enabling earlier identification of struggling students, and with SDG 10 (Reduced Inequalities) by explicitly accounting for socio-economic disadvantage rather than overlooking it [1-3, 9-11].

6. Brief description of Existing Research and Systems (300 words max)
7.

Most existing systems track only a single signal at a time. LMS dashboards typically flag risk based on login frequency or grade cut-offs, while the majority of research relies on either GPA and demographic data [1,4,5] or LMS clickstream data alone [4], with much of this work originating from Western contexts. Some studies have examined psychosocial factors [6-8] or financial and social factors [9-11], but these are generally investigated in isolation rather than integrated with academic or engagement data.

Several gaps emerge from this body of work. Most models only react once academic decline has already become apparent, and few combine more than one or two dimensions of risk [2,3]. Many also function as a "black box," producing a risk score without explaining the underlying reasoning [12-14]. Furthermore, environmental and financial factors common to the Sri Lankan context are rarely incorporated into these models [9-11]. This project seeks to address these gaps by constructing separate risk indices for each dimension and integrating them through an explainable AI layer.

8. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed system integrates four distinct risk dimensions rather than relying on a single measure, and provides an explanation for each prediction rather than producing a score alone, thereby addressing the gaps identified in Section 6.

Input: Academic data (GPA trend, attendance, marks); Engagement data (Moodle logins, time spent on resources, quiz behaviour); Psychosocial data (monthly survey covering stress, motivation, belongingness, and self-efficacy); Environmental data (income, commute time, internet/device access, part-time employment).

Process: For each data stream, several candidate models (logistic regression, random forest, gradient boosting) are compared and the best-performing one is selected, producing a score ranging from 0 to 100. SHAP is then applied to identify which factors contributed most to each component score. A Fusion Framework combines the four scores, and a final Explainable AI layer identifies the overall most significant drivers.

Output: a Final Attrition Risk score accompanied by a plain-language explanation (e.g., "high risk: low engagement and financial stress").

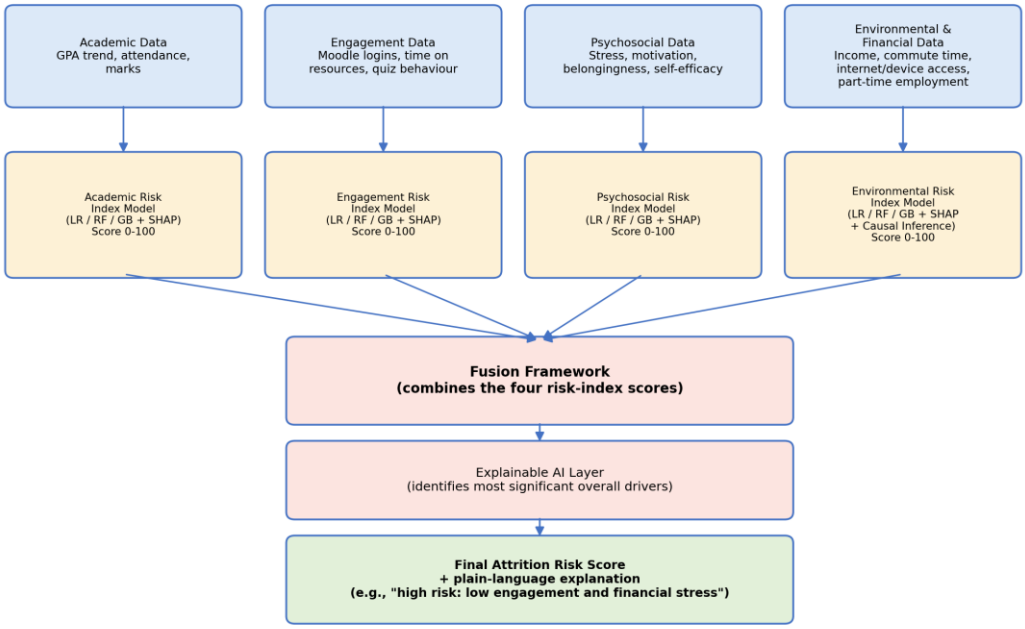


Figure 1: Conceptual diagram showing the fusion of the four risk dimensions (Academic, Engagement, Psychosocial, Environmental & Financial) into a single explainable attrition risk score.

9. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Domain expertise The required domain expertise will be developed through consultation with the research supervisor and co-supervisor, supported by a comprehensive review of literature on student attrition prediction, learning analytics, and educational data mining. The psychosocial component will utilise established and validated measurement scales for constructs such as stress, motivation, self-efficacy, and sense of belonging, ensuring the reliability of collected data. Guidance from academic experts and, where appropriate, university counselling staff will be sought during instrument selection. Explainable Artificial Intelligence (XAI) techniques, including SHAP and LIME, will be studied through academic publications, official documentation, and implementation on public datasets before integration into the proposed framework.

Technical skills: The technical knowledge required for implementing the framework will be acquired through structured online learning, official software documentation, and continuous supervisor guidance. Team members will develop competencies in machine learning, data preprocessing, multimodal data fusion, and Explainable AI using resources such as Coursera, Kaggle Learn, Scikit-learn, XGBoost, and SHAP documentation. Regular code reviews and collaborative development sessions will ensure methodological consistency across all four research components prior to system integration.

Data collection: Academic data, including GPA trends, examination results, and attendance records, will be obtained from the university's student records office following formal approval and anonymisation. Student engagement information will be collected from Moodle or Learning Management System (LMS) activity logs for participants providing informed consent. Psychosocial and environmental data will be gathered through a single anonymous, closed-ended questionnaire combining two response formats. Psychosocial constructs (stress, motivation, self-efficacy, and belongingness) will be measured using validated five-point Likert-scale instruments — the Perceived Stress Scale (PSS-10; Cohen et al.), the Academic Motivation Scale (AMS; Vallerand et al.), the General Self-Efficacy Scale (GSE; Schwarzer & Jerusalem), and the Psychological Sense of School Membership scale (PSSM; Goodenow) — each scored from "strongly disagree" to "strongly agree." Environmental and financial constructs (household income, commute time, internet/device access, and part-time employment) will be captured using closed-ended multiple-choice and categorical items with fixed response options (e.g., income bracket ranges, commute-time bands, and yes/no access and employment items). No open-ended or free-text questions will be used, so that all numerical responses can be directly incorporated into the proposed risk-index models. If institutional data cannot be obtained within the project timeline, a realistic synthetic dataset based on published Sri Lankan higher education studies will be used to validate the proposed framework.

Ethical clearance: Ethical clearance will be obtained from the University's Institutional Research Board (IRB) prior to data collection. Participation will be voluntary, with informed consent obtained from all participants. All data will be anonymised, securely stored, and accessible only to the research team and supervisors, ensuring compliance with institutional ethical standards and accepted research data management practices.

10. Objectives and Novelty

Main Objective To build and test a system that predicts student attrition risk by combining academic, engagement, psychosocial, and environmental data into one explainable risk score, so that at-risk students can be identified earlier and more accurately than with academic performance alone.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Gunasekara V.S.H. (IT23164680)	Academic Intelligence Framework	Collect, preprocess, and clean academic data, including GPA trends, examination marks, attendance records, and assignment submission delays. Structure the data as ordered semester-wise sequences per student rather than static feature vectors. Develop a sequence-aware model — an LSTM or GRU network — to learn academic trajectory patterns directly from the semester sequences, capturing gradual decline that single-point features would miss. Compare this sequence model's performance against a Gradient Boosting baseline trained on hand-engineered trend features (GPA slope, rolling average, attendance variance) to justify the choice of approach. Apply SHAP (or a suitable sequence-model explainability technique, such as attention-weight analysis or SHAP on the trend-feature baseline) to identify the academic factors contributing most significantly to attrition risk. Finally, generate and validate the Academic Risk Index (0–100), representing each student's level of academic attrition risk.	Academic trajectory analysis is incorporated to evaluate changes in students' academic performance across multiple semesters rather than relying on a single GPA snapshot. By capturing longitudinal performance trends, the proposed approach enables the early identification of gradual academic decline, providing a more comprehensive and reliable indicator of attrition risk than isolated academic measurements.

Vimukthi H.A.V.C. (IT23191242)	Multimodal Student Engagement Intelligence	Collect and preprocess Moodle/LMS activity logs, identifying key engagement indicators, including login frequency, time spent on learning resources, quiz-attempt behaviour, and assignment-interaction patterns. Apply clustering or a Hidden Markov Model (HMM) to the session-level behavioural data to discover distinct engagement states (e.g., consistent, declining, sporadic) rather than relying on raw activity counts. Use these discovered engagement states, alongside aggregated behavioural features, as inputs to a Gradient Boosting (XGBoost/LightGBM) classifier, and compare against Logistic Regression and Random Forest to select the best-performing model. Apply SHAP to identify the engagement factors and behavioural states contributing most to attrition risk, and develop the Multimodal Engagement Index (0–100).	The proposed component combines multiple behavioural indicators, including LMS activity, time spent on learning resources, quiz behaviour, and assignment interactions, rather than relying solely on login counts, enabling a more comprehensive and accurate assessment of student engagement.
Samarakoon S.M.S.R.K. (IT23279148)	Psychosocial Intelligence Framework	Design and administer a psychosocial survey using validated scales to measure stress, motivation, belongingness, self-efficacy, and peer support. Evaluate the reliability of the collected data using Cronbach's Alpha, preprocess the data, and develop machine learning models including Random Forest and XGBoost to predict psychosocial risk levels. Apply SHAP-based explainability to identify the psychosocial factors contributing most significantly to the prediction. Finally, generate and validate the Psychosocial Risk Index (PRI).	The proposed component integrates validated psychometric assessment with explainable machine learning to develop a Psychosocial Risk Index tailored to the Sri Lankan higher education context. The generated index contributes to a multi-factor student attrition prediction framework by incorporating psychosocial factors alongside academic, engagement, and environmental dimensions.
Bombugala B.A.C.D. (IT23445154)	Environmental & Financial Causal Risk	Collect and preprocess environmental data, including household income, commute time, internet and device access, and part-time employment status. Develop a Gradient Boosting (XGBoost) classifier for predictive risk	Unlike existing models, which are largely based on developed-country datasets and treat environmental factors in isolation, the proposed

	Intelligence Framework	scoring, compared against Logistic Regression and Random Forest baselines, and select the best-performing model. Apply SHAP to identify the environmental factors most predictive of attrition risk. As a separate analytical layer, apply causal inference techniques — Propensity Score Matching (PSM) or Inverse Probability Weighting (IPW) — to estimate the effects of key environmental factors on attrition risk while controlling for confounding variables, and use a Causal Forest (via the EconML library) to capture heterogeneous treatment effects, examining how environmental factors interact with academic performance and student engagement. Finally, generate and validate the Environmental Risk Index (0–100), incorporating both the predictive score and the causal effect estimates.	component applies causal inference techniques to estimate the effects of individual environmental factors on attrition risk while examining their interaction with academic performance and student engagement.

11. <u>This section is to be completed by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	√	No	

b) The proposed topic exhibits novelty.

Yes	√	No	

c) The student group can successfully execute the proposed project.

Yes	√	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	√	No	

e) Data required for the project is available.

Yes		No	√

f) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor*	Ms.	Chathurika	Koswatte	
Co-Supervisor*	Ms.	Anuththara	Ekanayake	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

References

1. Haarika, S., & Srinivas, K. (2022). Student dropout prediction using machine learning techniques. International Journal of Intelligent Systems and Applications in Engineering.

2. Predicting Student Dropout based on Machine Learning and Deep Learning: A Systematic Review. (2023). EAI Endorsed Transactions on Scalable Information Systems.

3. Machine Learning and Deep Learning for Dropout Prediction in Higher Education: A Review. (2026). MDPI.

4. Predicting student dropouts with machine learning: An empirical study in Finnish higher education. (2024). ScienceDirect.

5. Supervised machine learning algorithms for predicting student dropout and academic success: a comparative study. (2024). Discover Artificial Intelligence, Springer Nature.

6. Psychosocial factors impacting the college adjustment of undergraduate students: A scoping review.

7. Examining Factors Associated with Attrition, Strategies for Retention Among Undergraduate Nursing Students, and Identified Research Gaps: A Scoping Review. (2024/2025). PMC.

8. Developing and validating a university needs instrument to measure the psychosocial needs of university students. PMC.

9. Addressing the Complex Challenges Faced by Developing Countries in Achieving Educational Equity for Disadvantaged Youth in Tertiary Education: A Case Study of Sri Lanka. (2025).

10. Determinants of school drop-out among adolescents in rural Sri Lanka. (2024). ScienceDirect.

11. School Dropouts in Sri Lanka: A Sociological Analysis.

12. A hybrid framework of statistical, machine learning, and explainable AI methods for school dropout prediction. (2025). PLOS One.

13. Interpretable Dropout Prediction: Towards XAI-Based Personalized Intervention. (2023). International Journal of Artificial Intelligence in Education, Springer Nature.

14. Student Dropout Prediction Using Ensemble Learning with SHAP-Based Explainable AI Analysis (EASE-Predict). (2025).

IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

This part is to be filled by the Topic Screening Panel members.

Project ID: ……

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes (should be followed up by the supervisor)*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

The Review Panel Details

Member's Name	Signature	Date

*__Important__:

1. According to the comments given by the panel, make the necessary modifications and get approval from the **Supervisor** or the **Same Panel**.
2. If the project topic is rejected, identify a new topic, and follow the same procedure until the topic is approved by the assessment panel.